

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ**

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0515

Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

**Assessment Tool Weightage:
10%**

QUESTIONS:50

**Total Marks 50
Annexure A
MCQ**

Duration :60 Minutes

Code of Conduct:

I PAVAN JANA (192472313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Pavan j*

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Image Normalization

An image pixel value $r = 150$ is normalized to the range $[0,1]$ from 8-bit data. What is the normalized value?

Answer: C) 0.59

2. Image Storage Calculation

A grayscale image of size 1024×512 is stored using 8 bits per pixel. What is the storage requirement?

Answer: B) 512 KB

3. Edge Detection – Gradient Magnitude

Given Sobel gradients G_x and G_y , what is the gradient magnitude?

Answer: A) 10

4. Edge Direction Calculation

For the same gradients G_x and G_y , what is the edge direction?

Answer: C) 53.1°

5. Harris Corner Detection Response

Given $\det(M) = 200$, $\text{trace}(M) = 30$, $k = 0.04$, what is the corner response R ?

Answer: B) 164

6. Harris Corner Classification

If the Harris response $R = +250$, the pixel is classified as:

Answer: C) Corner

7. Hough Transform – Line Representation

A line detected at $\theta = 90^\circ$ and $\rho = 50$ corresponds to:

Answer: B) Vertical line at $x = 50$

8. SIFT Scale

An image is scaled by a factor of 2. If the original keypoint scale is $\sigma = 1.6$, new scale is:

Answer: C) 3.2

9. PCA Dimensionality Reduction

A feature vector has 100 dimensions. PCA retains 95% variance using 20 components. What is the dimensionality after PCA?

Answer: B) 20

10. PCA Covariance Matrix Size

For a dataset with 50 features, what is the size of the covariance matrix?

Answer: C) 50×50

11. Image Quantization

A grayscale image is quantized from 256 levels to 32 levels. What is the number of bits per pixel after quantization?

Answer: C) 5

12. Image Storage Reduction

An RGB image of size 512×512 uses 8 bits per channel. It is converted to grayscale with 8 bits per pixel. What is the percentage reduction in storage?

Answer: C) 66.7%

13. Sobel Edge Response

Given Sobel kernel response values , , what is the gradient magnitude?

Answer: C) 13

14. Gradient Direction Calculation

For , , the gradient direction is closest to:

Answer: B) 157°

15. Harris Corner Detection Response

Given $\det(M) = 400$, $\text{trace}(M) = 40$, $k = 0.06$, compute the Harris response R.

Answer: B) 304

16. Harris Corner Decision

For a pixel, eigenvalues of matrix M are $\lambda_1 = 120$, $\lambda_2 = 130$. This pixel is classified as:

Answer: C) Corner

17. Hough Transform Voting

A point contributes votes to 180 values of θ in Hough space. What is the shape of its representation?

Answer: C) A sinusoidal curve

18. Hough Line Detection

A detected line has parameters $\rho = 0$, $\theta = 45^\circ$. Which line is represented?

Answer: C) Diagonal line through origin

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of 4×4 histograms, each with 8 bins. What is the descriptor dimensionality?

Answer: C) 128

20. PCA Eigenvalue Selection

Eigenvalues of covariance matrix are [40, 25, 15, 10, 5, 5]. How many principal components are needed to retain 80% variance?

Answer: B) 3

21. Preprocessing – Normalization Scenario

A surveillance camera outputs 8-bit grayscale images. A pixel has value 204. What is the normalized value?

Answer: C) 0.80

22. Image Representation Scenario

A medical image of size 2048×1024 is stored using 16 bits per pixel. What is the memory required?

Answer: B) 4 MB

23. Edge Detection Scenario

At a pixel location, Sobel gradients are , . What is the edge magnitude?

Answer: A) 15

24. Edge Orientation Scenario

Using the same gradients , , what is the edge direction?

Answer: C) 53.1°

25. Harris Corner Detection Scenario

For a region, $\det(M) = 300$, $\text{trace}(M) = 25$, $k = 0.04$. What is the Harris response R ?

Answer: C) 275

26. Harris Corner Interpretation Scenario

If the Harris response R is large and positive, the pixel corresponds to:

Answer: C) Corner

27. Hough Transform Scenario

A point in image space votes for all possible θ values in Hough space. What shape does this vote form?

Answer: C) A sinusoidal curve

28. Hough Line Detection Scenario

A detected line has parameters $\rho = 40$, $\theta = 90^\circ$. Which line is represented?

Answer: A) Vertical line $x = 40$

29. SIFT Feature Scenario

An image is scaled by a factor of 0.5. A SIFT keypoint detected at scale $\sigma = 2.0$ originally will now appear at:

Answer: A) 1.0

30. PCA Dimensionality Reduction Scenario

A feature vector has 200 dimensions. After PCA, 30 principal components retain 92% variance. What is the new feature dimension?

Answer: A) 30

31. Preprocessing – Bit Depth Conversion

A grayscale image originally stored in 12-bit format is converted to 8-bit. An original pixel value is 3000. What is the approximate 8-bit value?

Answer: C) 187

32. Image Representation – Sampling Scenario

A camera samples an image at 600×400 pixels. The image is down-sampled by a factor of 2 in both directions. What is the new image size?

Answer: B) 300×200

33. Edge Detection – Sobel Response

At a pixel, Sobel gradients are $G_x = 15$, $G_y = 2$. What is the edge magnitude?

Answer: C) 17

34. Edge Orientation Scenario

For $G_x = 15$, $G_y = 2$, the gradient direction is closest to:

Answer: C) 118°

35. Harris Corner Detection – Computation

Given $\det(M) = 500$, $\text{trace}(M) = 35$, $k = 0.05$. What is the Harris response R?

Answer: A) 439

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents: Answer:

B) Edge

37. Hough Transform – Line Interpretation

A line detected with parameters $\rho = 0$, $\theta = 90^\circ$ corresponds to: Answer:

A) Vertical line at $x = 0$

38. SIFT Descriptor Scenario

Each SIFT descriptor contains 4×4 blocks with 8 orientation bins each. How many values describe one keypoint?

Answer: 128

¹ Note: 128 is the correct answer, but it was missing from the options you provided.

39. PCA Variance Retention

Eigenvalues of covariance matrix are [50, 30, 10, 5, 5]. How many principal components are needed to retain 90% variance?

Answer: C) 3

40. PCA Projection Scenario

Original feature vector dimension = 120. PCA reduces it to 25 components. What is the reduction percentage?

Answer: C) 79%

41. Quantization Error Analysis

An 8-bit grayscale image is quantized to 64 levels. Original pixel value = 150. What is the maximum possible quantization error?

Answer: B) ± 2

42. Image Representation Storage Comparison

Image A is 1024×1024 grayscale (8-bit). Image B is 1024×1024 RGB (8-bit per channel). What is the storage ratio (B:A)?

Answer: C) 3:1

43. Sobel Edge Strength Comparison

Pixel-1: . Pixel-2: . Which pixel corresponds to a stronger edge?

Answer: B) Pixel-2

44. Harris Corner Response Comparison

Two regions have the following Harris parameters:

Region det(M) trace(M)

R1 300 25

R2 350 40

Given $k = 0.04$, which region is more likely a corner?

Answer: B) R2

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45. Harris Corner vs Edge Decision

Eigenvalues of matrix M are $\lambda_1 = 120$, $\lambda_2 = 10$. What does this pixel represent?

Answer: B) Edge

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at $(\rho = 60, \theta = 45^\circ)$. What does this intersection indicate?

Answer: C) A line in image space

47. Hough Transform Parameter Effect

Increasing the θ resolution from 1° to 0.5° results in:

Answer: C) Higher line detection accuracy

48. SIFT Scale Analysis

An object appears at scale $\sigma = 1.6$ in Image-1. In Image-2, the object is scaled by $1.5\times$.

At what scale should the corresponding SIFT keypoint appear?

Answer: C) 2.4

49. PCA Variance Contribution Analysis

Eigenvalues of covariance matrix are [60, 20, 10, 5, 5]. How many principal components are needed to retain 90% variance?

Answer: C) 3

50. PCA Dimensionality Reduction Impact

Original feature dimension = 150. After PCA = 30 components. What is the percentage reduction?

Answer: C) 80

